

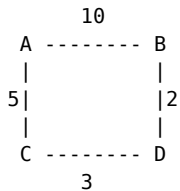
OSPF: Intuition to Protocol Architecture

From First Principles to Cisco CLI Implementation

Source: ChatGPT Shared Session (6ab9d884-3fa0-83ee-a21a-44aede996a47)

1. What Problem Does OSPF Solve?

Consider a four-router network:



Packets from **A** to **D** have multiple candidate paths:

- $A \rightarrow B \rightarrow D$: Cost = $10 + 2 = 12$
- $A \rightarrow C \rightarrow D$: Cost = $5 + 3 = 8$

OSPF's objective: **Learn network topology, then compute shortest, loop-free path to every destination.**

2. Link-State vs. Distance-Vector

- **Distance-Vector (e.g., RIP):** Routers ask neighbors: "What destinations do you know, and how far are they?" Routers lack global topological visibility.
- **Link-State (OSPF):** Routers flood local link conditions: "Here are my connected links and their costs." Every router in an area reconstructs the identical complete network map (LSDB), then runs Dijkstra's algorithm locally.

3. Core OSPF Processing Pipeline

```
Discover Neighbors (Hello) --> Establish Adjacency (2-Way / Full)
--> Exchange Link-State Info (DBD, LSR, LSU)
--> Synchronized LSDB --> Run Dijkstra (SPF)
--> Best Paths to Routing Table --> Forward Packets
```

4. Metric: OSPF Cost

OSPF does not use hop counts. Metric is **Cost**:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

- Default Reference Bandwidth: 100 Mbps (10^8 bps).
- 10 Mbps link $\Rightarrow \frac{100}{10} = 10$.
- 100 Mbps link $\Rightarrow \frac{100}{100} = 1$.
- **Modern caveat:** Links ≥ 100 Mbps (FastEthernet, Gigabit, 10G) default to cost 1 unless reference bandwidth is raised using `auto-cost reference-bandwidth 10000` (10 Gbps).

5. Interior Gateway Protocol (IGP)

OSPF routes interior traffic within a single Autonomous System (AS). Exterior protocols like BGP handle routing between independent Autonomous Systems across the Internet.

6. Hierarchical Area Design

In flat networks with thousands of routers, frequent link flaps trigger domain-wide SPF recalculations. OSPF introduces **Areas** to partition topologies:

- **Area 0 (Backbone Area):** Mandatory central transit area. All non-backbone areas connect to Area 0.
- **Area Border Router (ABR):** Router connected to Area 0 and one or more non-backbone areas. Generates Type 3 Summary LSAs.
- **Autonomous System Boundary Router (ASBR):** Router connecting OSPF to external routing domains (BGP, static, EIGRP). Generates Type 5 External LSAs.
- **Internal Router:** Router whose interfaces all belong to a single area.

7. The Five OSPF Packet Types

All OSPF packets encapsulate directly into IP protocol 89:

1. **Type 1 (Hello):** Discovers neighbors, maintains keepalives, negotiates parameters, and elects DR/BDR.
2. **Type 2 (Database Description - DBD):** Summarizes LSDB headers so neighbors identify missing entries.
3. **Type 3 (Link-State Request - LSR):** Requests full details for specific LSAs missing from the LSDB.
4. **Type 4 (Link-State Update - LSU):** Transports requested or updated LSAs (the "envelope" carrying LSA "documents").
5. **Type 5 (Link-State Acknowledgment - LSAck):** Acknowledges receipt of LSU packets to ensure reliability.

8. Router ID (RID) Selection

Every router requires a unique 32-bit identifier:

1. Explicit configuration: `router-id <x.x.x.x>`.
2. Highest IP on any active loopback interface.
3. Highest IP on any active physical interface.

9. Neighbor Adjacency States

Routers transition through 7 states to form adjacency:

1. **Down:** No Hello packets heard from neighbor.
2. **Init:** Hello received, but local RID not yet seen in neighbor's active neighbor list.
3. **2-Way:** Bidirectional communication established. DR/BDR election occurs on multi-access links. Non-DR routers remain in 2-Way/DROTHER with one another.
4. **ExStart:** Negotiates Master/Slave relationship and initial sequence number. Higher RID becomes Master.
5. **Exchange:** Routers exchange DBD summary packets.
6. **Loading:** Routers send LSRs for missing/outdated LSAs and process incoming LSUs.
7. **Full:** Link-state databases are fully synchronized.

10. Multi-Access Networks: DR and BDR

To avoid $\frac{N(N-1)}{2}$ adjacency relationships on broadcast links (e.g. Ethernet):

- Routers elect a **Designated Router (DR)** and **Backup Designated Router (BDR)**.
- Non-DR routers (DROTHER) only form Full adjacencies with the DR and BDR.
- Multicast addresses: 224.0.0.5 (All OSPF routers) and 224.0.0.6 (All DR/BDR routers).

11. Core OSPF Tables

1. **Neighbor Table** (`show ip ospf neighbor`): "Who is directly connected?"
2. **Database Table** (`show ip ospf database`): "What does the full area topology look like?"
3. **Routing Table** (`show ip route ospf`): "What is the best forwarding path to each destination?"

12. LSA Types Summary

- **Type 1 (Router LSA):** Flooded by every router within its area; describes attached links and costs.
- **Type 2 (Network LSA):** Flooded by the DR on multi-access links; lists all connected routers.
- **Type 3 (Network Summary LSA):** Generated by ABRs; advertises inter-area routes (0 IA).
- **Type 4 (ASBR Summary LSA):** Generated by ABRs; advertises how to reach an ASBR.
- **Type 5 (AS External LSA):** Generated by ASBRs; advertises external routes redistributed into OSPF (0 E1/0 E2).
- **Type 7 (NSSA External LSA):** Used in Not-So-Stubby Areas; translated to Type 5 by ABRs when crossing into Area 0.

13. Adjacency Requirements (Mismatch Troubleshooting)

Adjacency fails if any of the following do not match across the link:

- Area ID
- Subnet and subnet mask
- Hello and Dead timers (default: 10s / 40s)
- Authentication credentials and type
- Area flags (Stub / NSSA)
- Interface MTU (MTU mismatch causes hang in ExStart/Exchange)
- Unique Router IDs

14. Cisco IOS Configuration & Verification

! Method 1: Global process network command

```
Router(config)# router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 10.1.2.0 0.0.0.255 area 0
```

! Method 2: Direct interface command

```
Router(config)# interface GigabitEthernet0/0
```

```
Router(config-if)# ip ospf 1 area 0
```

! Verification commands

```
show ip protocols          ! Routing protocol operational status
```

```
show ip ospf neighbor      ! Neighbor status and adjacency state
```

```
show ip ospf interface brief ! Configured OSPF interfaces, costs, and area IDs
```

```
show ip ospf database      ! LSDB contents and LSA classifications
```

```
show ip route ospf         ! SPF-calculated routes installed in RIB
```